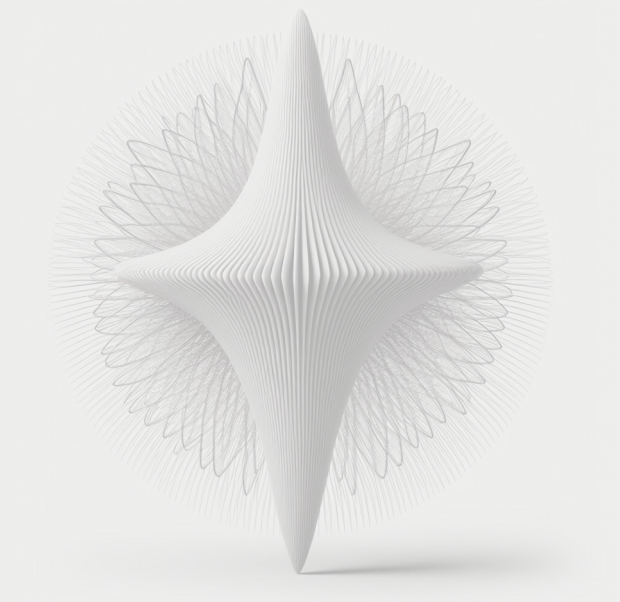


Signals and Systems

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Disclaimer: There could be errors and typos in these slides. Use with caution.

Laplace Transform

What if FT does not exist?

- How to analyze aperiodic signals $x(t)$ with infinite energy? Laplace Transform
  - Convert $x(t)$ to a finite energy signal: $x(t)e^{-\sigma t}$

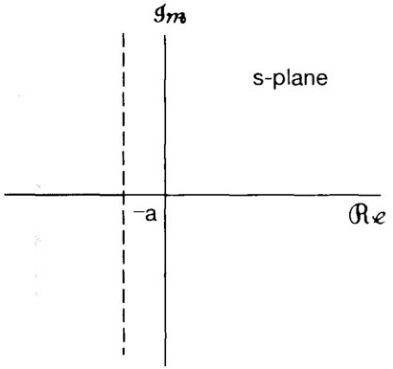
$$X(\sigma + j\omega) = \int_{-\infty}^{\infty} \left[x(t)e^{-\sigma t}\right] e^{-j\omega t} dt \qquad x(t)e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{j\omega t} d\omega$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt; \quad s = \sigma + j\omega \qquad x(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} X(s)e^{st} ds; \quad ds = j d\omega$$

  - Integration in the complex plane, along a line with real value = σ

Region of Convergence

- $X(s)$ exists only for those values of σ for which the FT$\{x(t)e^{-\sigma t}\}$ exists: ROC
  - If ROC includes σ = 0, then FT also exists
  - If σ is part of the ROC, then $\int |x(t)|\, e^{-\sigma t}\, dt < \infty$
  - Contains only strips parallel to jw axis

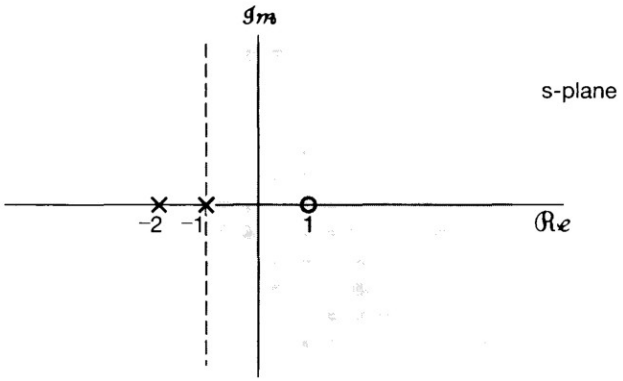


- If $x(t)$ is finite duration & bounded, then ROC is the entire s-plane
- If $x(t)$ is right sided and σ = a is in ROC, then σ > a is also in ROC
- If $x(t)$ is left sided and σ = a is in ROC, then σ < a is also in ROC

Laplace Transform Examples

- Find LT of $e^{-at}u(t)$ and $-e^{-at}u(-t)$

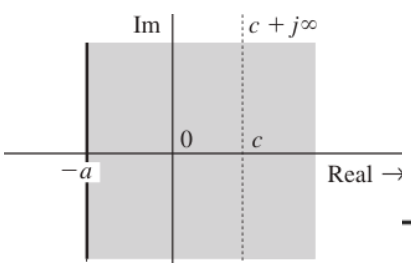
The Pole-Zero Plot

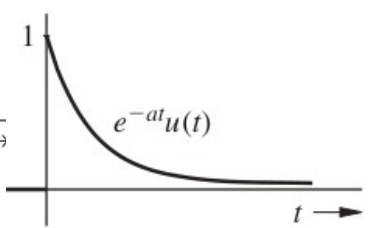


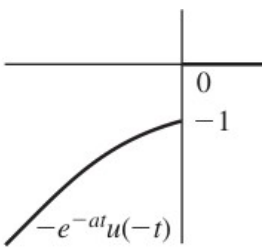
- For rational $X(s) = N(s) / D(s)$
  - Poles: roots of $D(s)$
    - Number of poles = degree of $D(s)$ polynomial
    - Order of pole = number of times that root is repeated in $D(s)$
  - Zeros: roots of $N(s)$
    - Number of zeros = degree of $N(s)$ polynomial
    - Order of zero = number of times that root is repeated in $N(s)$

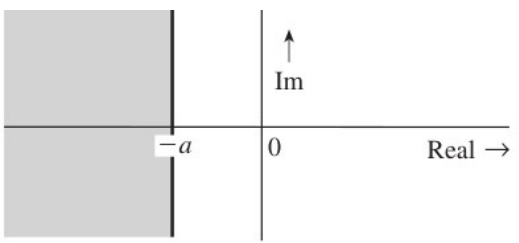
ROC of Rational Functions

- For rational LT,
  - There are no poles inside the ROC; poles bound the ROC
    - If $x(t)$ is right sided, then ROC is the all of the right of right-most pole
    - If $x(t)$ is left sided, then ROC is the all of the left of left-most pole
    - If $x(t)$ is two sided, then ROC is a strip between the poles



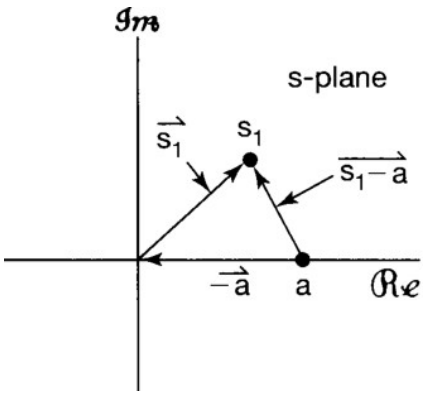






Geometric Interpretation of PZP

- $|X(s)|$ can be computed geometrically
  - Frequencies farther from the poles and closer to the zeros are attenuated
  - Frequencies closer to the poles and farther from the zeros are amplified



Bode Plot

- $|H(s)| = C\,\Pi|(s - a_i)| \,/\, \Pi|(s - b_i)|$
  - At $s = jw$, $|s - a_i| = |a_i|\sqrt{(w^2/a_i^2 + 1)}$
  - $|H(s)| = C' + 10\log(1 + w^2/a_i^2) - 10\log(1 + w^2/b_i^2)$
    - At $w = a_i$, $10\log(1 + w^2/a_i^2) = 3$ dB
    - At $w << a_i$, $10\log(1 + w^2/a_i^2) \approx 0$ dB
    - At $w >> a_i$, $10\log(1 + w^2/a_i^2) \approx 20\log(w) - 20\log(a_i)$
      - At, $w' = 10\,w$, $20\log(w') = 20$ (dB/decade) $+ 20\log(w)$

Laplace Transform - Properties

$x(t)$ – Time domain	$X(s)$ – Laplace domain	ROC	Poles & Zeros (for rational $X(s)$)
$x^*(t)$	$X^*(s^*)$	No change	Conjugated
$x(-t)$	$X(-s)$	Flips direction	Sign changes
$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	$R_1 \cap R_2$ (unless the sum is finite length)	Zeros change, Poles: $\{P_1\} \cup \{P_2\}$
$x(t - v)$	$X(s)e^{-jsv}$	No change	Not a rational form
$x(t)e^{vt}$	$X(s - v)$	$R + Re\{v\}$	$\{P\} + v$, $\{Z\} + v$
$x(ct)$	$X(s/c) / \lvert c \rvert$	cR: ROC is scaled $\lvert c \rvert > 1$: expands; $\lvert c \rvert < 1$: shrinks	$c\{P\}$, $c\{Z\}$

Convolution in Laplace Transform

Time domain	s - domain	ROC	Poles & zeros (for rational $X(s)$)
$x(t) * y(t)$	$X(s)Y(s)$	$R_1 \cap R_2$ (unless result is finite length)	$\{P_1\} \cup \{P_2\}$ $\{Z_1\} \cup \{Z_2\}$ Bode plots add up
$x(t)\, y(t)$	No relationship		
-	$X(s)*Y(s)$		

LT of Differentiation and Integrator

- LT$\{d^n x(t)/dt^n\} = s^n X(s)$, ROC is usually larger, another zero at origin

- LT$\{\int x(\tau)d\tau\} = X(s) / s$, ROC $\cap$ right half s-plane, another pole at origin

ILT of Differentiation

- $\mathrm{ILT}\{d^n X(s)/ds^n\} = (-t)^n\, x(t)$

- $\mathrm{ILT}\{(s + a)^{-n}\} = t^{n-1}e^{-at}u(t)/(n-1)!$

LT and LTI Systems

- Transfer function: $H(s) = Y(s) / X(s)$
  - This relationship true because e^{st} is an eigen function of LTI
    - But e^{st} do not form orthogonal basis
  - Transfer function becomes frequency response for $s = jw$

LT and LTI Systems

- Causal → ROC is right sided
  - Converse is true only for rational ($M < N$) transfer functions
- Stable → ROC includes s = 0
  - A causal system with rational transfer function is stable if and only if all poles are on the left half of the s-plane

Initial and Final Value Theorem

- For signals such that $x(t) = 0$ for $t < 0$,
  - Initial value theorem: $\lim_{s\to\infty} sX(s) = \lim_{t\to 0} x(t) = x(0)$
  - Final value theorem: $\lim_{s\to 0} sX(s) = \lim_{t\to\infty} x(t) = x(\infty)$

Example of Laplace Transforms

- Find the LT of $\delta(t)$, $\cos wt\ u(t)$, $e^{-at}\cos wt\ u(t)$

Example of Laplace Transforms

- Find LT of $e^{-|t|}$ and comment on causality and stability

Example of Laplace Transforms

- Comment on causality and stability of transfer function $e^{s} / s+1$ for Re{s} > -1

Example of Laplace Transforms

- Analyze the system $(s - 2)/(s + 2)(s - 5)$ for different possible ROCs

LT and RLC circuits

- Impedance in RLC circuits
  - Complex number
    - Z = a + b/s + cs
      - a = b = 0, c > 0: inductive
      - a = c = 0, b > 0: capacitive
      - b = c = 0, a > 0: resistive

$$v_R(t) = Ri(t) \qquad V_R(s) = RI(s)$$

$$v_C(t) = \frac{1}{C}\int_{-\infty}^{t} i(s)ds \qquad V_C(s) = \frac{I(s)}{Cs}$$

$$v_L(t) = L\frac{di(t)}{dt} \qquad V_L(s) = LsI(s)$$

Summary of Laplace Transform

Summary

- Laplace transform of x(t) is $X(s)$
  - Obtained from FT but making the signal absolutely integrable

$$X(\sigma + j\omega) = \int_{-\infty}^{\infty} \left[x(t)e^{-\sigma t}\right] e^{-j\omega t} dt \qquad x(t)e^{-\sigma t} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\sigma + j\omega) e^{j\omega t} d\omega$$

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st} dt; \quad s = \sigma + j\omega \qquad x(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} X(s) e^{st} ds; \quad ds = j d\omega$$

  - LT exists only for some values of s: ROC
- Converts differential equations to polynomials in s

Summary

ROC

- Contains only strips parallel to jw axis in the complex s-plane
- If ROC includes $\sigma = 0$, then FT exists and the system is stable
- If the signal is right sided, then its ROC is also right sided
- If the signal is left sided, then its ROC is also left sided
- If the signal is finite-length, then its ROC is the entire s-plane
- There are no poles inside the ROC

Summary

Rational Laplace Transforms

- Zeros: roots of numerator; poles: roots of denominator
  - If the signal is right sided, then its ROC is right of right most pole
  - If the signal is left sided, then its ROC is left of left most pole
  - If the signal is two-sided, then its ROC is a strip between two adjacent poles
- Pole-zero plot: around the pole: passband; around the zeros: stopband
- Bode plot: -20 dB/decade from every pole, +20dB/decade from every zero

Summary

$x(t)$ – Time domain	$X(s)$ – Laplace domain	ROC	Poles & Zeros (for rational $X(s)$)
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$x(ct)$	$X(s/c) / \lvert c \rvert$	cR: ROC is scaled $\lvert c \rvert > 1$: expands; $\lvert c \rvert < 1$: shrinks	$c\{P\}$, $c\{Z\}$

Summary

LTI Systems

- e^{st} are eigenfunctions of LTI systems, but not-orthogonal
  - Parseval's theorem: Not true for Laplace transform
- $\mathrm{LT}\{x(t) * y(t)\} = X(s)Y(s)$, ROC = $R_x \cap R_y$, $\{P_x\} \cup \{P_y\}$ and $\{Z_x\} \cup \{Z_y\}$
- Transfer function: Ratio of LT of output to LT of input $H(s) = Y(s)/X(s)$
  - Causality implies right sided ROC, converse is true ONLY for rational $H(s)$
  - A causal & rational $H(s)$ is stable iff all poles are on the left half of the s-plane

Summary

- Differentiation (HPF)
  - LT{dx(t)/dt} = sX(s); ROC is usually larger, another zero at origin
  - ILT{dX(s)/ds} = -t x(t)
- Integration (LPF)
  - LT{$\int$x(τ)dτ} = X(s)/s; ROC $\cap$ right half s-plane, add another pole at origin
- ODEs can be solved easily by partial fractions and LT{(a + s)$^{-n}$} = t^{n-1}e^{-at}u(t)/(n-1)!
- Initial/final value theorems: $x(0) = \lim_{s\to\infty} sX(s)$ $x(\infty) = \lim_{s\to 0} sX(s)$